Electrostatic Potential and Capacitance CHEAT SHEET

ELECTRIC POTENTIAL

$V = \frac{W}{q}$ energy per coulomb; 1 V = 1 J/C

2 μC across 3 cm of 1000 N/C: 60 μJ, so 30 V

only differences count; zero: far away, or the ground

the path doesn't matter (two roads, both 59.9 μJ)

point charge: $V = k\frac{Q}{r}$ (one over r, not r^2)

1 nC: 90 V at 10 cm, 45 V at 20 cm

CONSTANTS USED

$k = 8.99 \times 10^9$ N m^2/C^2; $\varepsilon_0 = 8.85 \times 10^{-12}$ $C^2/N\ m^2$

$e = 1.60 \times 10^{-19}$ C; $m_e = 9.11 \times 10^{-31}$ kg

air sparks at about 3×10^6 V/m (dry, sea level)

κ: air 1.0006, paper ≈ 3, glass 4 to 10, water ≈ 80

MANY CHARGES AND THE DIPOLE

many charges: $V = V_1 + V_2 + \ldots$ (numbers, with signs)

V = 0 does not mean E = 0 (1350 N/C there)

dipole, far: $V = \frac{kp\cos\theta}{r^2}$; 0 on the middle line

double r: a quarter (a charge: a half)

EQUIPOTENTIALS AND THE SLOPE

equipotential: one potential all over; no work along it

field lines cross them square on, pointing downhill

1 nC: 90 V at 10 cm, 60 V at 15 cm, 30 V at 30 cm

$E = -\frac{\Delta V}{\Delta r}$ 1 V/m = 1 N/C

30 V across 3 cm: 1000 V/m

POTENTIAL ENERGY

two charges: $U = k\frac{q_1 q_2}{r}$ (+ pushed together, − bound)

three charges: add every pair once

a charge in a field: U = qV

electron across 300 V: 1.0×10^7 m/s, any gap (sim.)

dipole: $U = -pE\cos\theta$ (lowest lined up)

EDGE TESTS

r grows: V → 0; a negative charge: a hollow

V the same everywhere: E = 0

dipole formula: 1 % off at 10a, 0.1 % at 30a

plates: rims bulge; too strong a field and air sparks

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CONDUCTORS

inside a conductor at rest: $E = 0$, one potential

charge on the surface; just outside $E = \frac{\sigma}{\varepsilon_0}$, square on

joined balls 10 cm, 2 cm, 12 nC: 10 and 2 nC, 899 V

the small one packs 5 × the charge per m^2: sharp points

a hollow conductor shields its inside (no field)

CAPACITORS

$C = \frac{Q}{V}$ charge per volt; $1\ F = 1\ C/V$

total charge zero: $+Q$ and $-Q$

parallel plates: $C = \frac{\varepsilon_0 A}{d}$

10 × 20 cm, 1 mm of air: 177 pF

160 μF with 1 mm of air: 18,000 m^2

DIELECTRICS

molecules stretch or turn: polarisation

field inside cut κ times; $C = \frac{\kappa \varepsilon_0 A}{d}$

battery off: Q stays, V falls κ times

battery on: V stays, Q rises κ times

mica holds 118 MV/m, air 3 MV/m

SERIES AND PARALLEL

series: the same Q; $\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2}$

2 and 3 μF on 10 V: 12 μC each, 6 V and 4 V

parallel: the same V; $C = C_1 + C_2$

ENERGY STORED

$U = \frac{1}{2}QV = \frac{1}{2}CV^2 = \frac{Q^2}{2C}$ (the triangle)

spoonfuls: 4 → 0.75, 1000 → 0.999 of $\frac{1}{2}CV^2$ (sim.)

energy density: $u = \frac{1}{2}\varepsilon_0 E^2$; air's limit ≈ 40 J/m^3

THE CAMERA FLASH

160 μF at 300 V: $Q = 0.048$ C; $U = 7.2$ J

in over ≈ 5 s: 1.4 W; out in 1 ms: 7200 W

oxide film $\kappa \approx 9$, 0.46 μm: foil ≈ 0.93 m^2

CHECK BEFORE YOU TRUST IT

before you trust an answer:

1 is it in V, J or F (and the right prefix)?

2 did you keep the signs of the charges?

3 in series, did you add the reciprocals?

PYTHON AND THE SIMULATIONS

Python: `V = 8.99e9 * Q / r`; `C = 8.85e-12 * A / d`

series: `1 / (1/C1 + 1/C2)`; `U = 0.5 * C * V**2`

equipotentials: step at right angles to E (RK4)

each simulation checked against its formula